Valerie Hetherington

Caltech Computer Science B.S 2024

Software Engineer | Zoot Enterprises — Aug 2025 – Present

- Worked across the application, synchronization, observability, configuration, and deployment layers of a distributed Java platform, supporting the backbone of financial decisioning and workflow automation for leading retailers and financial institutions
- Diagnosed performance and concurrency failures by instrumenting client-server synchronization paths and process-model execution flows, identifying event-processing bottlenecks and deadlock conditions that caused multi-second latency, hung threads, and service instability
- Enhanced system observability through analysis of logging infrastructure, thread-local context propagation, stack-trace generation, and connection-pool diagnostics using Log4j2, c3p0, and custom platform tooling
- Resolved client-impacting defects across desktop UI, workflow configuration, model synchronization, and deployment subsystems, improving reliability for credit decisioning and workflow automation applications
- Identified and drove resolution of a long-standing configuration portability issue by correcting XML serialization and schema-compliance defects that caused exported platform configurations to fail re-import, restoring a critical workflow for users migrating configurations between environments, including air-gapped deployments

Lead Software Engineer — Borys Lab, Montana State University / MonArk Quantum Foundry — Jan 2025 - Aug 2025

- Built backend infrastructure and data pipelines for an automated quantum materials research platform, integrating robotics, microscopy imaging, and sample metadata management using FastAPI, Supabase/Postgres, and Docker.
- Developed real-time image processing workflows that monitored microscopy data, performed AI-assisted flake detection and crystallite analysis, and uploaded extracted material characteristics for downstream research.
- Created researcher-facing tools for reviewing and correcting crystallite classifications, enabling collection of high-quality training data to improve computer vision model performance.

Software Engineering Intern at ActionTRAK — July 2022 - Nov 2022

- Fixed production bugs and added Java Swing UI components for an application managing product shipping and storage workflows.
- Developed database-integrated features, enhanced user interfaces, and resolved production defects delivered to end users

Embedded Systems Engineering Intern at Advanced Farm Technologies — Jun 2021 - Aug 2021

- Developed and debugged embedded C firmware for autonomous agricultural robotics systems, integrating ADCs, I2C peripherals, sensors, and interrupt-driven interfaces to improve strawberry harvesting throughput.
- Implemented temperature, tilt, and motion sensing capabilities using ADCs and IMUs, providing real-time telemetry and state information to downstream robotics control systems.
- Authored engineering documentation and debugging guides for custom silicon bring-up and J-Link-based development workflows.

Undergraduate Research Fellow Caltech Institute for Quantum Information and Matter (IQIM) — Jun 2020 - Sept 2020

- Implemented and evaluated classical versions of quantum simulation algorithms, including Trotter and qDrift, to study the tradeoffs between simulation accuracy and computational efficiency
- Designed and executed experiments comparing multiple qDrift implementations, validating prior research that a simplified formulation achieved performance comparable to the original algorithm.

Software Engineering Intern at Sellout Ticketing — Jun 2019 - Aug 2019

- Contributed to a full-stack ticketing platform by developing React/Apollo GraphQL user interfaces and backend functionality.
- Built SMS-based account verification and password recovery workflows using Twilio and Node.js, improving authentication and account recovery processes.

Projects - Github: https://github.com/skifast

- **MP3 Frame Counter (TypeScript)** Built an API that parses MPEG-1 Layer III frame headers byte by byte to count frames, with no parsing libraries
- **Durable Job Engine (Go, PostgreSQL)**: Job queue with worker leases, fencing tokens, idempotent submits, retries, and human approval; 20 database invariant checks, load and crash tests.
- **Problem Tracker (Python)** Built a coding-practice tracker that syncs solved LeetCode problems, removes duplicates, and schedules reviews with the SM-2 spaced-repetition algorithm.

Technical Skills

Languages & Engineering Practices: Java, Python, C/C++, JavaScript, SQL, Verilog, OCaml, Haskell, MATLAB, Object-Oriented Design, Go
Backend & Distributed Systems: FastAPI, Node.js, REST APIs, GraphQL, RabbitMQ, WebSockets, JDBC, MySQL, PostgreSQL, Supabase, Event-Driven Architectures, Client-Server Systems
Frontend & UI: React, Apollo GraphQL, Java Swing, HTML5, CSS3, Responsive Web Design
Infrastructure & DevOps: Docker, Kubernetes, Linux/Unix, Git, Maven, npm
Systems & Reliability: Concurrency, Performance Profiling, Log4j2, c3p0, XML/XSD Serialization, Configuration Management, Observability
Machine Learning & Computer Vision: Image Processing, AI Model Integration, CUDA, Real-Time Data Pipelines, Familiarity with training: Linear Regression, Lasso Regression, Ridge Regression, CNNs, Random Forests, Decision Tree, Markov Chains
Embedded Systems & Hardware: Embedded C, Verilog HDL, ADCs, I2C, IMUs, Interrupt-Driven Systems, J-Link Debugging, Sensor Integration
Scientific Computing: Quantum Algorithms (Trotter, qDrift), Numerical Computing, Linear Algebra, Simulation & Benchmarking

Awards and Honors

🎙️ TEDx Deerfield speaker: "Blockchain Beyond Bitcoin", February 2018
💣 Caltech Cannon Master: maintainer and launcher of a 1.7 ton, 130-year-old cannon for Caltech events, 2020-2022
⚽ Caltech Intercollegiate Women's Soccer - 2 Goalkeeping Records, 2020-2022
🏅 Awarded Semifinalist in U.S. Presidential Scholars Program, Spring 2019
🏅 Received award "for Excellence in Computer Science" from Deerfield Academy, May 2019